Equations and their graphs

1 The graph of the linear equation $y = 5x - 9$ is uniquely defined by a gradient of 5 and ___________ of -9. Tick **1** correct answer

- [ ] y-axis
- [ ] y coordinate
- [x] y intercept
- [ ] y solution

2 Which of these graphs will not form straight line when plotted? Tick **1** correct answer

- [ ] $y = 7x - 2$
- [ ] $y = 2 - 7x$
- [x] $y = x^2 - 7$
- [ ] $y = -2x - 7$
- [ ] $7x + 2y = 1$

3 Which of these will be y values if you populated this table for values for the linear equation $y = 7 - 2x$? Tick **3** correct answers

x	-3	-2	-1	0	1	2	3
y							

- [ ] -2
- [x] 1
- [ ] 2
- [x] 7
- [x] 9

4 From left to right how will the y values in this table be read if you complete the table for the linear equation $y = 4x - 9$? Tick **1** correct answer

x	-3	-2	-1	0	1	2	3
y							

- [ ] -3, -2, -1, 0, 1, 2, 3
- [ ] -17, -13, -9, -5, -1, 3, 7
- [ ] -21, -25, -29, -33, -37, -41, -45
- [x] -21, -17, -13, -9, -5, -1, 3
- [ ] 3, -1, -5, -9, -13, -17, -21

5 The linear graph $3x + 5y = 25$ is plotted. When the x coordinate is 0 the y coordinate is 5. Fill in the blank

6 The quadratic graph of $y = x^2$ is plotted. When the x coordinate is -5 the y coordinate is 25. Fill in the blank